Open-Vocabulary 3D Scene Graphs from Sparse RGB Views: Duplicate-Aware Fusion and a De-Circularized Benchmark

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Abstract. We study how to build and—crucially—how to *honestly evaluate* open-vocabulary 3D scene graphs from a handful of casually captured RGB views. Our pipeline keeps every heavy model frozen and uses no per-scene optimization: feed-forward VGGT geometry [26] turns sparse images into world-point maps, an open-vocabulary detector (OWLv2) [15] proposes labeled object boxes, and these are lifted to 3D and fused across views into a labeled object–relation graph. We make three points. First, replacing the common class-agnostic-mask + per-patch CLIP [10,20] front-end with a genuine open-vocabulary detector is what makes the labels real: on a desk scene the detector recovers the actual objects (keyboard, monitors, book) where the mask+CLIP pipeline returned room-scale category noise. Second, the dominant failure of the lifted system is *over-counting*—one physical object survives as several fused nodes—and a simple duplicate-aware fusion step (merge same-label nodes whose 3D boxes overlap) fixes it, improving object-label F1 at every view count and winning on 22–27 of the 28 benchmark scenes ($p < 0.001$). Third, a previously promising uncertainty-aware fusion gate gives *no* benefit once labels are real, across a full parameter sweep—a negative result that a circular evaluation had masked. To ground these claims we contribute an independent sparse-view benchmark on 28 TUM RGB-D [23] sequences (5 human-verified, 23 two-pass VLM-drafted) whose reference is authored from the frames rather than seeded from the model's own predictions, and we document the circular-pseudo-reference pitfall that inflates scores. We hope the benchmark and the de-circularization protocol prove useful beyond this system.

Keywords: 3D scene graphs · sparse views · open-vocabulary detection · evaluation methodology · benchmark · duplicate-aware fusion

1 Introduction

Reconstructing a scene and explaining a scene are related but different goals. Modern geometry pipelines can recover camera motion, depth, points, surfaces, or radiance fields, yet downstream applications often require object-level statements: which pixels across views depict the same object, where that object is

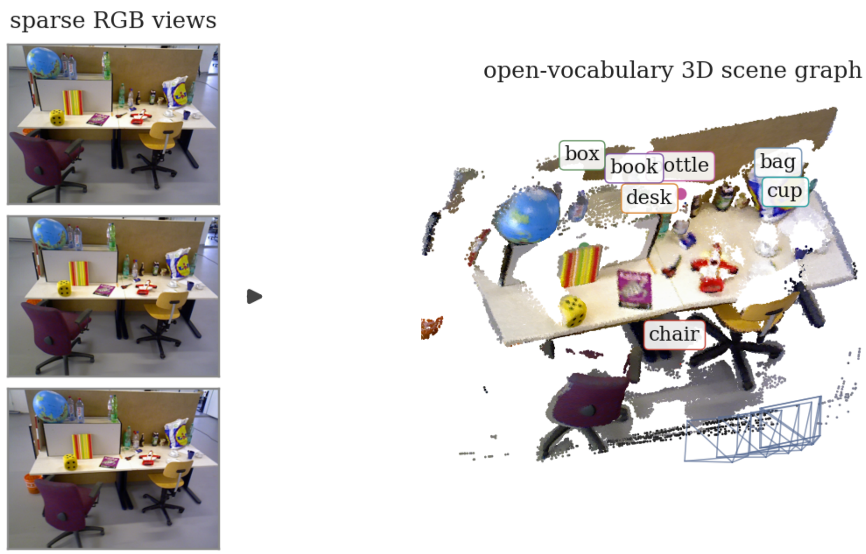

Fig. 1. From a handful of RGB views to an open-vocabulary 3D scene graph. **Left:** three of the input frames (TUM `freiburg3_long_office_household`). **Right:** the system lifts frozen-model detections into VGGT world-point geometry and fuses them across views into labeled 3D object nodes (colored markers) — a desk, books, bottles, a cup, a bag, a box, and two office chairs — with all heavy models frozen and no per-scene optimization. Best viewed in color.

in 3D, and how it is spatially related to other objects. A robot or mixed-reality system, for example, benefits more from a statement such as "the monitor is on the desk" than from an unlabeled collection of points. This need is especially acute when the input is small: a few casual RGB views rather than a calibrated RGB-D stream, dense video, or an already optimized map.

Classical structure-from-motion and SLAM systems remain powerful when overlap, calibration, and optimization time are available [21,2]. Recent feed-forward geometry models have changed the starting point by predicting camera parameters, depth, and point maps directly from one or more images [27,11,26]. At the same time, 2D foundation models provide strong open-vocabulary detections, language-aligned embeddings, class-agnostic masks, and robust self-supervised descriptors through detectors such as OWL-ViT and OWLv2, and through CLIP, SAM, and DINOv2 [16,15,20,10,17]. These components make sparse-view semantic 3D reasoning plausible, but they do not by themselves define an object-level graph. Per-image detections are local, repeated instances may have similar descriptors, and feed-forward geometry is least reliable near occlusions, thin structures, textureless regions, and large viewpoint changes.

We approach sparse-view scene graph construction as evidence-preserving fusion over lifted 2D object proposals, keeping all heavy perception models frozen and using no per-scene optimization. VGGT world-point maps serve as a 2D-to-3D lookup table; an open-vocabulary detector proposes labeled object boxes; these are lifted into candidate 3D objects, merged across views into a labeled object-relation graph, and annotated with spatial predicates (Figure 1). This design keeps the expensive perception stage fixed while making the graph reasoning stage explicit, auditable, and easy to ablate. It also differs from open-vocabulary 3D graph systems that assume posed RGB-D video, a dense point cloud, or a prebuilt map [4,6]: our target is a lower-input regime of sparse RGB views processed by feed-forward geometry.

A central lesson of this work is methodological. An earlier version of this system graded its predicted scene graphs against a pseudo-reference *seeded from the model's own dense-view output.* That protocol is circular: a fusion method that simply reproduces its dense-view graph scores 1.0 at the reference view by construction, and the apparent gains it reports need not survive an honest test. When we re-grade against an *independently authored*, human-verified reference — written from the frames rather than from any model prediction — two things change. First, a front-end that labeled image patches with CLIP turns out to recover almost none of the real objects: on a desk scene it returns room-scale category noise ("floor", "curtain", "bed") rather than the monitors, keyboard, and books that are actually present, and its headline label accuracy was an artifact of grading CLIP against CLIP. Replacing the patch-labeling front-end with a genuine open-vocabulary *detector* (OWLv2) makes the labels real: the detector recovers the actual objects on every frame. Second, and more cautionary, a previously promising uncertainty-aware fusion gate — which our earlier circular protocol reported as a sparse-view win — shows *no* benefit once labels are real, across a full weight sweep. We report this negative result as a feature, not a footnote: it is exactly the kind of conclusion a de-circularized evaluation is meant to surface.

With honest labels in hand, the dominant remaining error is over-counting: one physical object survives cross-view fusion as several same-label nodes (Figure 6). The fix is deliberately simple — a duplicate-aware merge of same-label nodes whose axis-aligned 3D boxes overlap. This step improves object-label F1 at every view count and is the only variant whose accuracy *rises* with more views, precisely because it converts the recall that extra views provide into precision rather than into duplicates. Notably, this is the opposite move from the uncertainty gate (which tends to over-split), and it explains why merging helps where the gate does not.

We evaluate on 28 public TUM RGB-D sequences under 3/5/8/10-view protocols [23], against an independently authored reference (5 scenes human-verified, 23 drafted by two independent passes). The duplicate-aware variant improves object-label F1 over the no-uncertainty graph-fusion baseline by +0.06 to +0.12 across view counts and wins on 22–27 of the 28 scenes ($p < 0.001$), while the recalibrated uncertainty gate is consistently *worse* than the baseline and an uninformed control. The gain holds on the 5 human-verified scenes alone and on both static and dynamic (moving-person) sequences.

Our contributions are:

- **A detector-grounded open-vocabulary sparse-view 3D scene-graph system.** An open-vocabulary detector (OWLv2) proposes labeled boxes, VGGT geometry lifts them to 3D, and a cross-view fusion stage produces a labeled object-relation graph with spatial predicates — all models frozen, no per-scene optimization. Detector-grounded labels recover real objects where a mask-plus-CLIP-patch front-end does not.
- **Duplicate-aware cross-view fusion.** A post-fusion merge of same-label nodes whose axis-aligned 3D boxes overlap (IoU $> \tau$), targeting the measured over-counting failure. It improves object-label F1 at every view count and is the only variant whose accuracy rises with more views.
- **An independent, human-verified sparse-view benchmark and a decircularization protocol on TUM RGB-D.** We document how a prediction-seeded pseudo-reference inflates scores (graph fusion scores 1.0 at the reference view by construction) and show that an independently authored reference removes the artifact.
- **A rigorous negative result on uncertainty-aware fusion.** Recalibrated and swept over weights, the uncertainty gate does not improve object accuracy on real labels; we analyze why (it lifts recall but over-splits, costing precision), which also explains why the opposite move — duplicate-aware merging — works.

2 Related Work

Feed-forward sparse-view geometry. Multi-view reconstruction and SLAM classically rely on correspondence, triangulation, bundle adjustment, and calibrated or densely overlapping observations. COLMAP remains a standard reference for incremental structure from motion [21], while ORB-SLAM3 demonstrates the strength of optimized visual and visual-inertial SLAM for coherent streams [2]. Feed-forward 3D models reduce this dependence on per-scene optimization. DUSt3R casts reconstruction as point-map prediction from unconstrained images [27]; MASt3R adds 3D-aware dense matching [11]; and VGGT predicts camera parameters, depth, point maps, confidence, and tracks from one or many views [26]. These models provide the geometric substrate for our work. We use VGGT as a frozen source of pixel-aligned 3D evidence, but our output is not a reconstruction: it is an object-relation graph grounded in that geometry.

Open-vocabulary detection as a labeling front-end. The semantics of a lifted scene graph are only as good as the 2D labels it ingests. A common recipe pairs class-agnostic masks from SAM [10] with per-region CLIP embeddings [20] matched against a text vocabulary, optionally backed by DINOv2 appearance descriptors [17]. This mask-then-embed route is attractive because every component is frozen, but it pushes the naming burden onto whole-image-trained CLIP applied to cropped regions, which tends to return scene- or room-scale category guesses rather than the individual objects present. A separate line of work performs *open-vocabulary detection* directly: the model localizes *and* names objects against an open text vocabulary in one pass. OWL-ViT attaches lightweight detection heads to a contrastively pre-trained image–text backbone and queries it with class-name embeddings [16]; OWLv2 scales this recipe with self-training on web image–text data and improves rare-category localization [15]; Grounding DINO marries a DINO detector with grounded language pre-training for open-set detection from text prompts [13]; and Detic broadens detector vocabularies by training on image-classification data [28]. We adopt an open-vocabulary detector (OWLv2) as the front-end of our system precisely because it emits localized, named instances, which is the property the lifting and fusion stages need. To our knowledge this family has been used mainly for 2D benchmarks; we study what it buys a sparse-view 3D scene-graph pipeline relative to the mask-then-embed alternative.

Open-vocabulary 3D semantics. In 3D, open-vocabulary scene understanding has been explored by lifting, distilling, or optimizing foundation-model features into point clouds, maps, masks, and neural scene representations. NeRF introduced continuous radiance fields for view synthesis [14]; LERF embeds language features into radiance fields for open-ended 3D queries [9]; and 3D Gaussian Splatting has become a common explicit substrate for fast scene representations [8]. OpenScene learns open-vocabulary point features aligned with image and text embeddings [18]; ConceptFusion builds queryable 3D maps by fusing pixel-aligned open-set features [6]; OpenMask3D aggregates CLIP features over class-agnostic 3D instance masks [24]; CUA-O3D studies cross-modal and uncertainty-aware agglomeration for open-vocabulary 3D scene understanding [12]; and OpenSplat3D performs open-vocabulary 3D instance segmentation in a Gaussian Splatting representation [19]. These methods motivate foundation-feature fusion in 3D. Our contribution is complementary: we construct explicit object-relation graphs from sparse RGB views without assuming a completed 3D map or optimizing a per-scene radiance field. Notably, uncertainty-aware aggregation has been proposed as a way to fuse multi-view evidence in this setting [12]; we revisit that idea on real detector labels under an independent reference and report a negative result (Section 5).

3D scene graphs. 3D scene graphs organize object instances and their relationships as structured representations for reasoning about scenes. Early work proposed unified graphs spanning objects, spaces, cameras, and semantic or spatial relationships [1]. 3DSSG introduced a large-scale dataset and learned prediction framework for semantic scene graphs from 3D indoor reconstructions [25]. RGB-D resources such as ScanNet further show the value of richly annotated reconstructed scenes for indoor understanding [3]. ConceptGraphs is closest to our output space: it builds open-vocabulary 3D scene graphs from posed RGB-D sequences by fusing 2D foundation-model outputs into object nodes and relations [4]. We share the object-relation motivation, but relax the input assumption. Instead of requiring a depth stream, posed RGB-D map, or completed 3D reconstruction, we lift sparse-view 2D proposals through feed-forward VGGT point maps and build object identities through cross-view geometric and semantic fusion. A recurring failure mode of any such lifting-and-fusion pipeline is *over-counting* – one physical object surviving as several fused nodes – which motivates the duplicate-aware merge we study.

Honest evaluation and reproducibility in 3D scene understanding. Evaluating open-vocabulary 3D scene graphs is itself hard: there is rarely a clean, instance-level ground truth for a handful of casual RGB views, so it is tempting to score a method against a reference derived from the system's own dense-view output. This couples the reference to the predictor and produces a *circular* evaluation in which a method can attain a perfect score at the reference view by construction, masking real differences between variants. Such reference–predictor coupling is a form of information leakage, which has been identified as a widespread driver of the reproducibility crisis in machine-learning-based science [7,5]. The broader 3D community has responded by curating carefully controlled benchmarks with independent ground truth – e.g. the Replica dataset of high-fidelity reconstructed indoor scenes [22] and the long-standing TUM RGB-D benchmark with externally measured trajectories [23]. In that spirit, we contribute an independent, human-verified sparse-view reference authored from the frames rather than seeded from the model's predictions, document the circular-pseudo-reference pitfall explicitly, and report which design choices actually help once the artifact is removed – including a negative result for uncertainty-aware fusion that the circular reference had hidden.

3 Method

3.1 Problem Setup

Given a sparse set of RGB views $\mathcal{I} = \{I_1, \ldots, I_V\}$, the goal is to produce an open-vocabulary 3D scene graph $\mathcal{G} = (\mathcal{O}, \mathcal{R})$. Each object node $o_i \in \mathcal{O}$ stores a 3D point set, centroid, an axis-aligned 3D bounding box, an open-vocabulary label, view support, a feature descriptor, and its source detections. Each relation edge $r_{ij} \in \mathcal{R}$ stores a spatial predicate between two object nodes. The method assumes neither depth images, externally supplied camera poses, nor a prebuilt mesh: it derives geometry and labeled object proposals from frozen feed-forward models, lifts the proposals to a common 3D frame, and fuses them into objects with no per-scene optimization. Figure 2 summarizes the data flow.

3.2 Overview

The pipeline has five stages, all built on frozen models. (i) VGGT predicts camera parameters, depth maps, world-point maps, and confidence maps for the sparse views [26]. (ii) An open-vocabulary *detector* (OWLv2) proposes labeled object boxes from a text query set, so each proposal already carries a real object label [15]. (iii) Each detection is lifted into the VGGT world-point map to form a 3D candidate fragment. (iv) Candidates from different views are fused

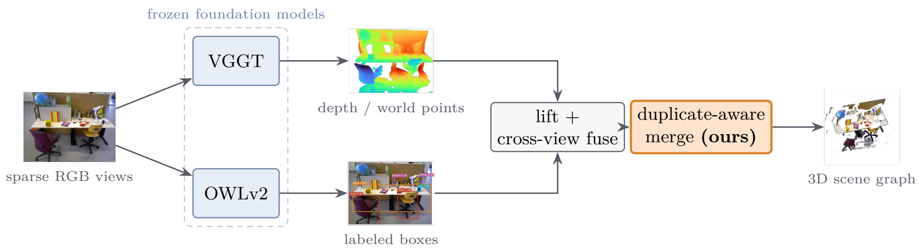

Fig. 2. The pipeline, on a real office scene. Sparse RGB views feed two *frozen* foundation models: VGGT produces world-point geometry (depth shown) and OWLv2 proposes open-vocabulary labeled boxes. Detections are lifted into the world-point map, fused across views into object nodes, and — our one addition — passed through a *duplicate-aware merge* (amber) that collapses same-label overlapping nodes, yielding the labeled 3D object–relation graph at right. All models are frozen; no per-scene optimization.

into object nodes, after which a duplicate-aware pass merges over-counted nodes that correspond to the same physical object. (v) Fused nodes are connected by rule-based spatial relations. The exported graph keeps the predicted labels and relations alongside the evidence that produced them, so it can be inspected or turned into annotation-review packets.

A central design change relative to a common open-vocabulary recipe deserves emphasis up front. A widely used front-end pairs class-agnostic masks [10] with per-region CLIP scoring [20] to label regions post hoc [6,4,24]. On blurry, low-resolution casual frames this front-end produced surface and background fragments that CLIP then labeled with room-scale categories, yielding labels disconnected from the actual objects (Section 4). We therefore replace it with a genuine open-vocabulary detector whose box *is* the labeled object hypothesis. Everything downstream of the front-end — lifting, fusion, and relations — is kept unchanged, which lets us attribute the resulting accuracy change to the front-end rather than to a re-tuned geometry stack.

3.3 Sparse-View Geometry

We run VGGT on the selected sparse views and store camera intrinsics, extrinsics, depth maps, world-point maps, and confidence maps [26]. The world-point map for each view acts as a 2D-to-3D lookup table: object proposals are lifted directly from image coordinates into a common 3D frame without first reconstructing a mesh, running bundle adjustment, or optimizing a radiance-field or Gaussian-Splatting representation [21,14,8]. For view v we denote the world-point map by $P_v(u) \in \mathbb{R}^3$ and the confidence map by $C_v(u)$ at image coordinate u.

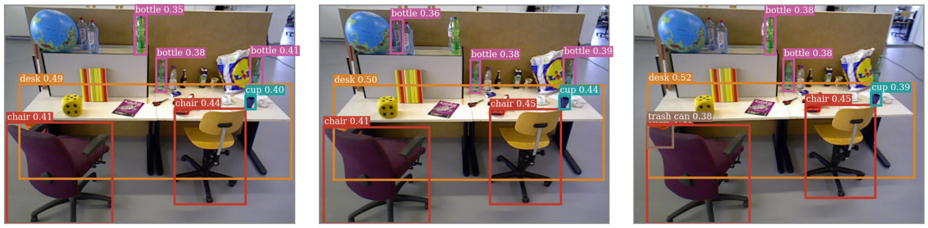

Fig. 3. The open-vocabulary detection front-end on real frames (three views of `freiburg3_long_office_household`). OWLv2, queried with the indoor vocabulary, returns named, boxed objects (desk, chair, bottle, cup, trash can) directly on each frame, with confidences — a genuine detector label per box. This is the property the lifting and fusion stages need; the common alternative (a post-hoc CLIP guess on a class-agnostic mask) returned room-scale category noise on these blurry frames rather than the objects actually present.

3.4 Open-Vocabulary Detection Front-End

For each view we run OWLv2 [15], an open-vocabulary detector that takes a set of free-form text queries and returns boxes with per-box class scores. The queries are the indoor object vocabulary of the benchmark, so the detector — not a post hoc patch-classification step — carries the open-vocabulary label. For detection k in view v we store its box, score $s_{v,k}$, and predicted label $\ell_{v,k}$. We use the `owlv2-base-patch16-ensemble` model with a detection threshold of 0.2 and per-class non-maximum suppression at box-IoU 0.5; the NMS mainly cleans the large surface classes (e.g. a desk) that otherwise accrue many overlapping boxes. We deliberately do *not* over-tighten the 2D threshold: a physical object may receive several overlapping boxes per frame, but overlapping boxes on the same object lift to nearly the same 3D points and are intended to collapse downstream. Optionally, each detection box prompts a frozen SAM [10] to produce a tight instance mask $M_{v,k}$ so that lifting selects only object pixels; without a mask, lifting falls back to the box rectangle. When language-aligned and self-supervised appearance descriptors are available, we additionally extract CLIP and DINOv2 features for each detection and store the normalized concatenation [20,17]; these are used only for fusion gating and provenance, never for labeling.

3.5 3D Lifting

Each detection's region (its instance mask $M_{v,k}$, or its box if no mask is available) is resized to the VGGT geometry resolution and used to select world points from the corresponding view; non-finite points and points below the confidence threshold τ_c are discarded:

$$X_{v,k} = \{P_v(u) \mid u \in M_{v,k},\ C_v(u) \geq \tau_c\}. \tag{1}$$

Detections with no valid lifted points are rejected. Each surviving candidate node stores its sampled 3D point set $X_{v,k}$, centroid $\mu_{v,k}$, an axis-aligned 3D bounding box, the detector label $\ell_{v,k}$ and score $s_{v,k}$, and (when available) the feature $f_{v,k}$.

3.6 Cross-View Fusion

Candidate nodes are fused into objects by building an association graph over the lifted detections. By default we disallow same-view association edges, so a connected component represents cross-view support for one object rather than a grouping of detections within a single image. An edge connects two candidates when their centroids are close in 3D and, when valid descriptors exist, their cosine similarity exceeds a feature threshold:

$$\|\mu_a - \mu_b\|_2 \leq \epsilon_g \quad \text{and} \quad \cos(f_a, f_b) \geq \epsilon_f. \tag{2}$$

The feature condition is omitted for candidates without valid descriptors. Connected components define fused object identities; for each component, points are pooled and sampled, descriptors are averaged and renormalized, and the node label is set by a score-weighted majority vote over the constituent detector labels $\{\ell_{v,k}\}$ (ties broken by count). This separates view-local evidence from object identity while preserving the detection provenance needed for inspection and annotation.

3.7 Duplicate-Aware Fusion

Cross-view fusion alone leaves the dominant precision failure unaddressed. Characterizing the fused system against an independent reference (Section 4) shows that the precision loss is almost entirely *over-counting of correctly found objects*, not mislabeling: as more views are added the detector finds more of the true objects (recall rises), but each physical object also accumulates more detections that fusion fails to collapse into a single node (precision falls). Two structural reasons make this expected. First, the detector over-fires, emitting several overlapping boxes per object per frame. Second, by design the association graph forbids same-view edges, so duplicate detections of one object within a frame can never be joined by cross-view fusion. The result is that a single desk or monitor survives as several fused nodes. Over-counting — not the labels themselves — is therefore the lever to pull, and the corrective move is *merging*, not splitting.

We add a single post-fusion pass, DEDUP, that merges fused nodes of the same label whose axis-aligned 3D boxes overlap. Let $\mathcal{O}$ be the fused nodes from Section 3.6. For a node o with point set X_o, let $B_o = [\, b_o^{\min},\, b_o^{\max}\,]$ be its axis-aligned 3D bounding box, with $b_o^{\min} = \min X_o$ and $b_o^{\max} = \max X_o$ taken coordinatewise. For two boxes the axis-aligned 3D intersection-over-union is

$$\mathrm{IoU}_{3D}(B_a, B_b) = \frac{\mathrm{vol}(B_a \cap B_b)}{\mathrm{vol}(B_a) + \mathrm{vol}(B_b) - \mathrm{vol}(B_a \cap B_b)}, \tag{3}$$

where $\mathrm{vol}(B_a \cap B_b) = \prod_{d=1}^{3} \max\!\big(0,\, \min(b_{a,d}^{\max}, b_{b,d}^{\max}) - \max(b_{a,d}^{\min}, b_{b,d}^{\min})\big)$. We partition $\mathcal{O}$ by label. Within each label group we form a merge graph whose vertices are the nodes and whose edges connect pairs with $\mathrm{IoU}_{3D}(B_a, B_b) > \tau$, and we take its connected components; each component is collapsed into one object by re-running the same pooling, descriptor averaging, and label vote used in fusion. Nodes with no label, and singleton groups, are passed through unchanged. Concretely, with $G_\ell = \{o \in \mathcal{O} \,:\, \mathrm{label}(o) = \ell\}$:

$$\textsc{Dedup}(\mathcal{O};\tau) = \bigcup_{\ell} \Big\{\, \textsc{Fuse}(c) \;:\; c \in \mathrm{CC}\big(G_\ell,\ \{(a,b) : \mathrm{IoU}_{3D}(B_a, B_b) > \tau\}\big) \Big\}, \tag{4}$$

where $\mathrm{CC}(\cdot)$ denotes connected components and $\textsc{Fuse}(\cdot)$ re-fuses a component into a single node (the identity on singletons). Restricting merges to same-label nodes keeps genuinely distinct objects of different classes separate, and requiring box overlap keeps spatially separated instances of the *same* class separate. The default threshold is $\tau = 0.1$; the step is robust across the low-IoU range and the IoU criterion alone outperforms adding a centroid-distance term (Section 4). We use 3D-box IoU rather than mask or point-set overlap because the lifted boxes are exactly the geometry already attached to each node, making the test parameter-free beyond τ. The one structural cost is that closely packed same-class instances (e.g. two monitors on one desk) can have overlapping boxes and be over-merged; we quantify this trade-off and discuss appearance-based instance splitting as the natural remedy in Sections 4 and 6.

3.8 Spatial Relations

Spatial relations are inferred from fused centroids using distance and relative-position rules. The relation vocabulary includes *near*, *left_of*, *right_of*, *above*, *below*, *in_front_of*, and *behind*. A *near* relation is emitted for object pairs within a distance threshold; otherwise the vertical displacement or the dominant horizontal coordinate determines the directional predicate, with an optional maximum relation distance suppressing very distant pairs. The exported graph preserves both the symbolic fields and the numeric support, so the same object identities, point sets, labels, and provenance can be reused by manual review or by a future learned relation head.

3.9 An Ablated Variant: Uncertainty-Aware Fusion

A natural hypothesis — and the headline of an earlier version of this system — is that fusion should be *uncertainty-aware*: each lifted candidate can be assigned an uncertainty

$$U_{v,k} = \lambda\,(1 - \bar{C}_{v,k}) + (1-\lambda)\,\mathrm{clip}\Big(\tfrac{\sigma(X_{v,k})}{s_0}, 0, 1\Big)\,, \tag{5}$$

combining VGGT point confidence $\bar{C}_{v,k}$ with the spatial spread $\sigma(X_{v,k})$ of the lifted points (in our runs $\lambda = 0.7$, $s_0 = 1$), and the fusion gate is then *tightened* for uncertain pairs — effectively splitting ambiguous candidates apart. We retain this as an ablated variant rather than as part of the core method, because (i) the failure analysis above shows the system over-counts, so the productive move is to merge more, not to split more, and a gate that splits uncertain pairs acts on the

Table 1. TUM RGB-D sequences prepared for sparse-view benchmarking. Each sequence contains 10 selected RGB frames and is evaluated with 3, 5, 8, and 10 input views.

Scene ID	Source sequence	Frames	View counts
`tum_rgbd_freiburg1_room`	`freiburg1_room`	10	3/5/8/10
`tum_rgbd_freiburg1_desk`	`freiburg1_desk`	10	3/5/8/10
`tum_rgbd_freiburg1_desk2`	`freiburg1_desk2`	10	3/5/8/10
`tum_rgbd_freiburg2_xyz`	`freiburg2_xyz`	10	3/5/8/10
`tum_rgbd_freiburg3_long_office_household`	`freiburg3_long_office_household`	10	3/5/8/10

wrong axis; and (ii) once labels are real and the evaluation is de-circularized, the variant gives no benefit over plain cross-view fusion across a full weight sweep (Section 4). We report it as a deliberate, rigorous negative result. The uncertainty score itself is still exported with each node, since it is potentially useful for the opposite operation — guiding an appearance-based *split* of over-merged groups — which we leave to future work (Section 6).

4 Experiments

We evaluate two questions that the reframed thesis turns on: (i) what actually helps when building open-vocabulary 3D scene graphs from a handful of RGB views, and (ii) how to measure that honestly. The second question is not incidental. The system we study was originally reported as a sparse-view win for uncertainty-aware fusion; that conclusion was an artifact of a circular evaluation, and exposing it required first building a reference that the system could not grade itself against. We therefore describe the benchmark and its de-circularization in detail before reporting the variant comparison (Section 5).

4.1 Benchmark Scenes and Sparse-View Protocol

We benchmark on public TUM RGB-D indoor sequences [23]. The benchmark spans **28 object-rich sequences** across `freiburg1`–`freiburg3`: the five paper-subset scenes in Table 1 (which carry the human-verified gold reference) plus 23 additional sequences; the full list is in `configs/datasets/tum_rgbd_expanded.json`. Each sequence contributes ten selected RGB frames (stride 10). Although the source benchmark is RGB-D, our pipeline consumes only RGB; we use this dataset because it provides public, recognizable indoor scenes with reproducible sequence identifiers.

For each scene we evaluate four input-view counts, $N \in \{3, 5, 8, 10\}$, giving a 28-scene by four-view-count matrix (112 graph outputs per variant). The benchmark runner creates one output directory per (scene, N) and executes the full pipeline along an identical code path: VGGT geometry, OWLv2 detection, descriptor extraction, 3D lifting, cross-view fusion, open-vocabulary labeling, and metric export. The driver is `scripts/run_owlv2_benchmark.sh`. Reporting accuracy as a function of N lets us read off the sparse-view behavior of each variant directly, rather than a single dense-view number.

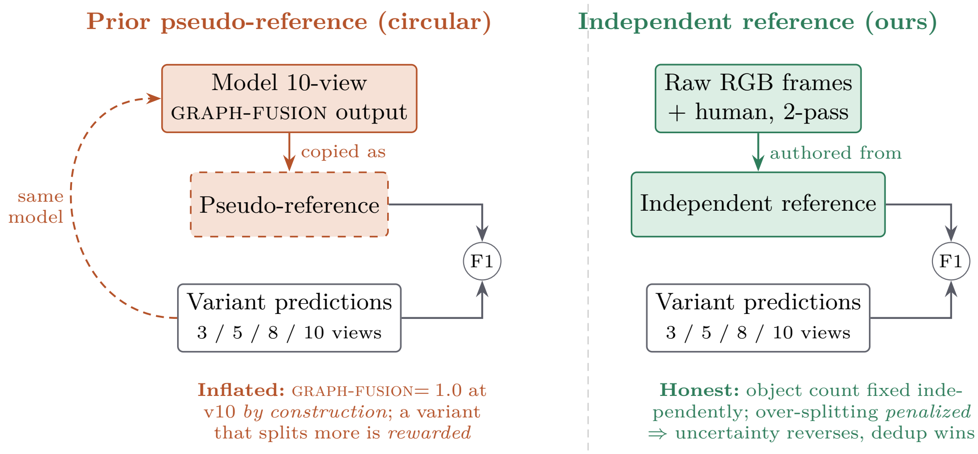

Fig. 4. De-circularizing the evaluation. **Left:** the earlier protocol seeded the reference from the model's own ten-view GRAPH-FUSION prediction, so that variant scores 1.0 at ten views *by construction*, and a variant that keeps more proposals unmerged (the uncertainty gate) is rewarded for reproducing the reference's own fragments—an apparent sparse-view win. **Right:** our reference is authored from the raw RGB frames, independently of any model output. The true object count is fixed, over-splitting is penalized, and both findings emerge: the uncertainty gate reverses to a loss and duplicate-aware fusion wins.

4.2 An Independently-Authored Reference

The central evaluation problem in this setting is that there is no ground-truth open-vocabulary object inventory for TUM RGB-D scenes, so a reference must be constructed. The tempting shortcut — and the one the original version of this work took — is to seed the reference from the system's own dense-view prediction (the 10-view graph) and then lightly check it. We argue this is unsound and replace it with a reference authored *from the frames*.

Why the prediction-seeded reference is circular. If the reference object multiset is initialized from a variant's own 10-view output (Figure 4, left), that variant scores precision = recall = 1.0 at the 10-view "reference" point *by construction*, and every sparser run is graded for how faithfully it reproduces the same variant's splitting and labeling pattern rather than for agreement with the scene. A method that keeps more uncertain proposals unmerged then looks better simply because it reproduces more of the fragments the reference inherited from the seeding variant. Under exactly this prediction-seeded reference, the uncertainty-aware variant appeared to be a sparse-view win (+0.040/+0.021 at N=3/5) and the fusion baseline hit 1.0 at N=10 by construction. Both effects are evaluation artifacts, not properties of the scenes.

Construction of the independent reference. We instead enumerate, for each scene, an object multiset built only from the ten raw RGB frames and *indepen-*

dent of the detection/descriptor/geometry/fusion stack under test. Each scene is drafted in one pass and then re-examined in a second, adversarial pass (an explicit attempt to find missed or spurious objects) over the same frames. For the five paper-subset scenes the two-pass draft is additionally **human-verified** against the frames, removing ambiguous detections (e.g. a window glimpsed in `room/desk/desk2`; cup/bottle/picture in `desk2`); these five form the gold core (`configs/evaluation/independent_labels.json`). The 23 additional scenes use the same two-pass draft *without* human verification (`independent_labels_expanded.json`) and serve as scale corroboration; we show both main findings also hold on the human-verified five alone. Per-scene provenance is logged in `docs/independent_reference_worklist.md`. Because this reference fixes the true object inventory without reference to any variant's output, no variant can score 1.0 at N=10 by construction, and over-splitting is correctly penalized rather than rewarded. The reference is a scene-level multiset over the ten frames; sparse-view predictions are recall-limited against it by construction, which depresses absolute F1 but applies uniformly across variants and so leaves the comparison fair.

4.3 Variants Compared

We hold the front-end (OWLv2 detections lifted by VGGT geometry) and the reference fixed, and compare seven variants that differ only in how per-view detections become 3D nodes and how those nodes are fused across views:

- **2d-only** — per-view OWLv2 labels pooled without 3D lifting or cross-view fusion (an image-level upper reference on labels alone).
- **geometry-only** — detections lifted to 3D via VGGT and pooled, with no semantic cross-view association.
- **semantic-lifting** — detections lifted and associated by appearance descriptors without the geometric fusion gate.
- **fixed-shrink (uninformed control)** — the fusion gate is modulated by a fixed, scene-agnostic shrink factor; an uncertainty-blind control that isolates whether the *uncertainty signal*, rather than merely the act of tightening the gate, carries any benefit.
- **graph-fusion (baseline)** — the standard cross-view fusion of Section 3 with no uncertainty modulation and no duplicate handling.
- **proposed (uncertainty)** — graph-fusion with the rank-normalized uncertainty-aware merge gate (default weight w=0.3); evaluated as an ablation rather than the headline method, and additionally swept over $w \in \{0.1, 0.2, 0.3, 0.5, 0.8\}$.
- **graph-fusion-dedup (ours)** — graph-fusion followed by a single post-fusion duplicate-instance merge: same-label fused nodes whose axis-aligned 3D boxes overlap (IoU $> \tau$, default τ=0.1) are unioned and re-fused (`merge_duplicate_instances` in `graph_builder.py`). This targets the measured over-counting failure (Section 5) and is the opposite move from the uncertainty gate, which *tightens* merging.

The pairing is deliberate: *fixed-shrink* versus *proposed* asks whether the uncertainty signal is informative beyond an uninformed gate adjustment, and *graph-fusion* versus *graph-fusion-dedup* asks whether merging *more* (rather than less) is the productive direction.

4.4 Metric

Our primary metric is `object_label_f1`: the multiset precision, recall, and F1 of fused-node labels against the independent reference multiset, averaged over the 28 scenes (5 human-verified, 23 two-pass VLM-drafted). Treating labels as a multiset (rather than a set) makes the metric sensitive to over-counting — a scene with one physical monitor reported as four fused monitor nodes is penalized on precision — which is exactly the failure mode the reframed contribution targets. Per-scene wins/losses are reported alongside the means, and significance is summarized with a two-sided sign-test, which over the 28 scenes reaches $p < 0.001$ for both main findings. Evaluation is run by `scripts/evaluate_sparse_view_annotations.py` and aggregated by `scripts/aggregate_variant_f1.py`; raw per-(scene, N, variant) scores are in `results/benchmark_owlv2_expanded/variant_independent_metrics.csv`.

4.5 Implementation Details

All heavy models are frozen and run off-the-shelf; there is no per-scene optimization or fine-tuning. The open-vocabulary detector is OWLv2 (`google/owlv2-base-patch16-ensemble`) [15], the successor to OWL-ViT [16], with a detection score threshold of 0.2 and per-class non-maximum suppression at IoU 0.5; detection boxes prompt SAM [10] for masks. 3D geometry is the feed-forward VGGT transformer [26], consuming RGB frames only. Appearance descriptors for cross-view association use CLIP [20] and DINOv2 [17]. VGGT geometry runs on GPU; detection, lifting, fusion, and evaluation run on CPU. We fix all thresholds across scenes and view counts (no per-scene tuning), and the duplicate-aware merge uses a single global IoU threshold τ=0.1 that we show is robust across $\tau \in [0, 0.2]$ (Section 5).

4.6 Gold Core, Drafted Corroboration, and Dynamic Stress

The 28-scene benchmark pairs a small *human-verified* core (the five paper-subset scenes) with 23 two-pass VLM-drafted scenes for statistical power. Both main findings reach $p < 0.001$ over the 28 scenes and reproduce on the five human-verified scenes alone (Section 5), so they do not depend on the drafted labels; fully human-verifying the 23 is the natural next step. Six sequences are *dynamic* (a moving person: `desk_with_person`, `sitting_*`, `walking_*`), which violates VGGT's static multi-view assumption; we report them as a stress subset. The duplicate-aware gain holds in both groups — +0.082 to +0.171 on the 22 static scenes and +0.060 to +0.116 on the 6 dynamic scenes — so the contribution is robust to scene dynamics as well as to scale.

5 Results

We evaluate on an independent sparse-view reference (Section 4): a per-scene object multiset enumerated directly from the RGB frames, authored independently of the SAM, CLIP, VGGT [26], and fusion stages. The reference covers **28 TUM RGB-D** [23] sequences: the **5 paper-subset scenes are human-verified** (the gold core), and the remaining **23 are drafted by two independent passes** (draft + adversarial verification) over the frames, pending human verification. The primary metric is *object-label F1* (`object_label_f1`): multiset precision/recall/F1 of fused-node labels against the reference multiset, reported as the mean over scenes at 3/5/8/10 input views. All numbers are measured against this independent reference unless stated otherwise; the contrast with the earlier prediction-seeded pseudo-reference is in Section 5.6.

5.1 Variant comparison on the independent reference

Table 2 compares seven variants: four lifting/fusion baselines (2D-ONLY, GEOMETRY-ONLY, SEMANTIC-LIFTING, and the no-uncertainty GRAPH-FUSION), an uninformed FIXED-SHRINK control that applies a constant merge-gate shrink with no per-node signal, the uncertainty-aware PROPOSED variant (rank-normalized, weight 0.3), and our duplicate-aware GRAPH-FUSION-DEDUP. On real, detector-grounded labels, GRAPH-FUSION is the strongest prior fusion variant, and object-label F1 *falls* as views are added for every cross-view fusion variant (SEMANTIC-LIFTING, FIXED-SHRINK, GRAPH-FUSION, PROPOSED); the non-fusion references (2D-ONLY, GEOMETRY-ONLY) plateau. More views supply more detections of the same physical object, which cross-view fusion does not collapse, so precision (and net F1) degrade against a fixed reference. GRAPH-FUSION-DEDUP is the only variant whose accuracy *rises* toward the dense end and is best at every view count (Figure 5).

5.2 Duplicate-aware fusion: the positive result

Duplicate-aware fusion adds a single post-fusion step: same-label fused nodes whose axis-aligned 3D boxes overlap (IoU $> \tau$, default τ=0.1) are merged and re-fused. Table 3 reports the gain over both the GRAPH-FUSION baseline and the uninformed FIXED-SHRINK control. Dedup improves object-label F1 at every view count by +0.057 to +0.122 over GRAPH-FUSION and by +0.065 to +0.154 over FIXED-SHRINK; the gain *grows* with views, as expected for a step that removes the duplicate detections that accumulate as more frames are seen. It wins on 22–27 of 28 scenes at every view count (two-sided sign test $p < 0.001$). The improvement holds on both the 22 static scenes (+0.082 to +0.171) and the 6 dynamic scenes with a moving person (+0.060 to +0.116), and on the 5 human-verified gold scenes alone ($+0.068/+0.089/+0.145/+0.146$), so it is not an artifact of the VLM-drafted references.

Table 2. Object-label F1 on the independent reference (mean over 28 TUM RGB-D sequences; 5 human-verified + 23 two-pass VLM-drafted, full multiset). Best per column in bold. FIXED-SHRINK is an uninformed control; PROPOSED is the rank-normalized uncertainty-aware variant (w=0.3). Accuracy falls with more views for every cross-view fusion variant except duplicate-aware fusion, which is best throughout.

Variant	v3	v5	v8	v10
2D-ONLY	0.5120	0.5003	0.4867	0.4753
GEOMETRY-ONLY	0.4617	0.4605	0.4594	0.4493
SEMANTIC-LIFTING	0.3752	0.2881	0.2123	0.1797
FIXED-SHRINK (control)	0.5048	0.4715	0.4448	0.4189
GRAPH-FUSION (baseline)	0.5129	0.4895	0.4664	0.4502
PROPOSED (uncertainty, w=0.3)	0.4801	0.4370	0.3841	0.3544
graph-fusion-dedup (ours)	**0.5696**	**0.5723**	**0.5860**	**0.5725**

Table 3. Duplicate-aware fusion versus the GRAPH-FUSION baseline and the uninformed FIXED-SHRINK control. Δ is the object-label F1 improvement; W/L/T counts are per-scene wins/losses/ties for dedup over GRAPH-FUSION (n = 28, sign test p < 0.001 at every view).

	v3	v5	v8	v10
Δ dedup − GRAPH-FUSION	+0.057	+0.083	+0.120	+0.122
Δ dedup − FIXED-SHRINK	+0.065	+0.101	+0.141	+0.154
Per-scene W/L/T (vs GRAPH-FUSION)	22/1/5	25/0/3	27/0/1	26/1/1

5.3 Uncertainty-aware fusion: a negative result

We treat the uncertainty-aware fusion gate — the headline of an earlier framing of this work — as an ablation. Table 4 reports its delta against GRAPH-FUSION. The recalibrated PROPOSED variant is *worse* at every view count (−0.033 to −0.096) and loses on essentially every scene (26–27 of 28 at v5–v10; sign test $p < 0.001$). A sweep over the uncertainty weight $w \in \{0.1, 0.2, 0.3, 0.5, 0.8\}$ does not rescue it: *no* weight in $[0.1, 0.8]$ beats GRAPH-FUSION at any view count, with larger weights worse at the dense end. The result is robust to reference filtering: dropping `unknown object` (which the open-vocabulary detector never predicts) or the "stuff" classes (wall/floor/ceiling) raises every variant's F1 but leaves the ranking unchanged.

The mechanism is informative rather than a mere null. The uncertainty modulation does exactly what it was designed to do — it lifts recall by declining to merge uncertain proposals (full-multiset recall, mean over scenes: PROPOSED 0.516/0.556/0.596/0.602 vs. GRAPH-FUSION 0.510/0.540/0.581/0.587) — but the same behavior fragments correct objects and costs more precision than the re-

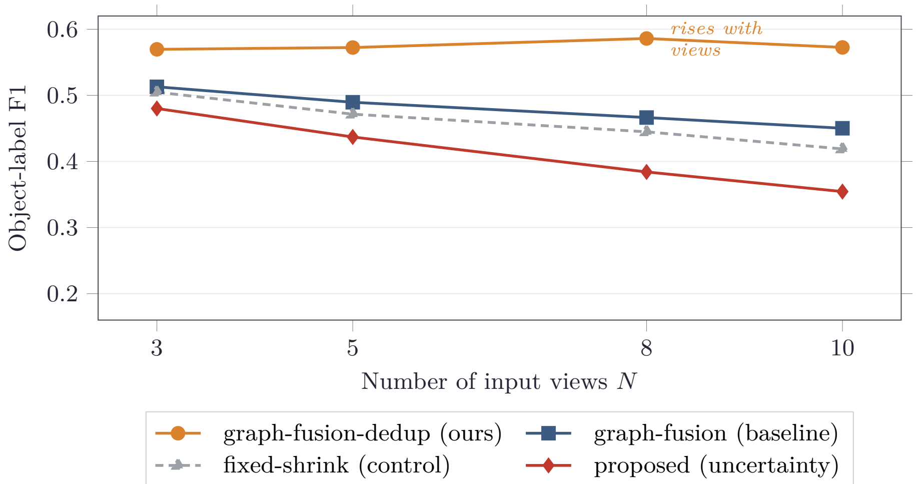

Fig. 5. Object-label F1 vs. number of input views on the independent 28-scene reference. Every prior cross-view fusion variant *degrades* as views are added—more frames yield more un-collapsed duplicate detections, costing precision. Duplicate-aware fusion (ours, amber) is the only variant whose accuracy *rises* toward the dense end and is best at every view count, while the uncertainty-aware gate (brick) sits below even the uninformed control. Values match Table 2.

call it buys (precision: PROPOSED 0.478/0.388/0.299/0.266 vs. GRAPH-FUSION 0.556/0.495/0.420/0.399). Against a fixed-count reference the net F1 is lower. Crucially, this is the *opposite* axis to duplicate-aware fusion: the uncertainty gate *tightens* merging and splits more, whereas the system's measured weakness (Section 5.4) is over-counting, which calls for the opposite move — merging more. The same per-class evidence explains why uncertainty fails and why dedup succeeds.

5.4 Per-class characterization: precision-up, recall-down

Table 5 reports the micro-averaged precision/recall/F1 of GRAPH-FUSION per view count (dropping `unknown object`, which the detector never emits). The signature is clear: recall rises from 0.700 at three views to 0.802 at ten, while precision falls from 0.519 to 0.331, so net F1 is best at the sparse end (0.596 at v3, 0.468 at v10). The system finds more true objects with more frames but emits more duplicate detections, and the precision loss dominates.

Per class (at v8), this splits into three behaviors. A group is *found reliably but over-counted* — recall near 1.0 while precision collapses: `desk` (0.15 precision), `wall` (0.26), `floor` (0.27), `keyboard` (0.39), `monitor` (0.43), `chair` (0.45). These are always detected, but the 3D fusion does not collapse the several detections of one physical object, making them the dominant precision sink. A second group is *clean wins* with high precision and recall (`book`, `bottle`, `picture`, `person`,

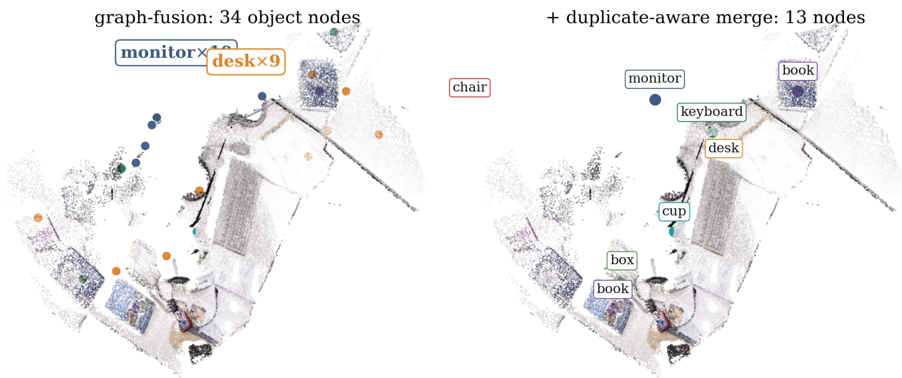

Fig. 6. Duplicate-aware fusion on the real `freiburg1_desk` point cloud (ten views; the cloud is desaturated so the object nodes stand out). **Left:** cross-view GRAPH-FUSION leaves the single monitor as ten nodes and the desk as nine — the over-counting that dominates the precision loss. **Right:** the duplicate-aware merge collapses same-label overlapping nodes into one object each ($34 \to 13$ nodes), recovering a clean labeled scene graph. (Merging is shown by 3D proximity for this illustration; the benchmark metric uses 3D-box IoU, Section 4.3.)

`cup`): distinct, well-separated objects that fuse correctly. A third group is *under-detected*: `cabinet` (recall 0.27), where the detector under-fires on blurry, low-resolution TUM frames. The dominant error is thus over-counting of correctly-found objects, not mislabeling — precisely what duplicate-aware fusion targets (Figure 7).

Table 6 confirms the mechanism at v8: dedup sharply raises precision on the over-counted classes (`desk` $0.15 \to 0.45$, `floor` $0.27 \to 0.71$, `monitor` $0.43 \to 0.64$, `cabinet` $0.36 \to 0.67$, `chair` $0.45 \to 0.71$) while leaving recall unchanged for almost every class. The only recall cost is on closely-packed same-class instances — `desk` and `bottle` lose -0.04 recall, as adjacent same-class objects with overlapping axis-aligned 3D boxes are merged. The net is a large F1 gain because the precision recovery dwarfs the recall cost; separating closely-packed same-class instances by appearance is left to future work (Section 6).

5.5 Robustness of the duplicate-aware merge

The dedup gain is robust to the IoU threshold. Table 7 sweeps $\tau \in \{0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.5\}$: the improvement over GRAPH-FUSION stays in the $+0.075$ to $+0.118$ range (average Δ) across the entire low-IoU band $\tau \in [0, 0.2]$ and only fades once the threshold is strict enough to stop merging duplicates ($\tau \geq 0.5$, $+0.030$). The wired default τ=0.1 is within noise of the optimum and slightly more recall-conservative than "any overlap." Adding a centroid-distance OR-term on top of IoU does not help, so 3D-box IoU alone is the right and simplest criterion.

Table 4. Uncertainty-aware fusion is a negative result. Δ PROPOSED − GRAPH-FUSION in object-label F1 (rank-normalized, w=0.3); W/L/T is per-scene over $n = 28$. The variant is worse at every view count ($p < 0.001$), and a weight sweep over $[0.1, 0.8]$ contains no setting that beats the baseline.

	v3	v5	v8	v10
Δ PROPOSED − GRAPH-FUSION	−0.033	−0.053	−0.082	−0.096
Per-scene W/L/T	2/22/4	0/26/2	0/26/2	0/27/1

Table 5. Per-class characterization of GRAPH-FUSION (micro-averaged over 28 scenes, dropping `unknown object`). Recall rises and precision falls as views are added; F1 is best at the sparse end. The precision loss is over-counting of correctly-found objects, not mislabeling.

Views	Precision	Recall	F1
3	0.519	0.700	**0.596**
5	0.429	0.739	0.543
8	0.361	0.794	0.496
10	0.331	0.802	0.468

5.6 De-circularization: the earlier pseudo-reference was misleading

The two findings above only become visible under an independent reference. The earlier evaluation of this system used a prediction-seeded pseudo-reference, drafted from the model's own ten-view GRAPH-FUSION output. Because that reference encoded GRAPH-FUSION's own splitting pattern, two artifacts followed. First, GRAPH-FUSION scored 1.0 at ten views *by construction* — the reference *was* its own prediction, so the dense-view row measured reference construction rather than accuracy. Second, the uncertainty-aware PROPOSED variant, which keeps more uncertain proposals unmerged and therefore splits more, reproduced more of the same fragments the circular reference contained, and so appeared to be a sparse-view *win* (+0.040 at v3, +0.021 at v5). The independent reference fixes the true object count: over-splitting is now correctly penalized, the uncertainty result reverses sign (Table 4), and the dense-view 1.0 artifact disappears (F1 instead falls with views, Table 5). We document this pitfall because a prediction-seeded reference is an easy and silent way to inflate scores in 3D scene understanding, and because it is what masked both of this paper's main empirical findings.

6 Discussion

Three findings of this paper pull in the same direction, and it is worth stating plainly why. On an independent, human-verified reference, a one-line duplicate-aware merge improves open-vocabulary object-label F1 at every view count and

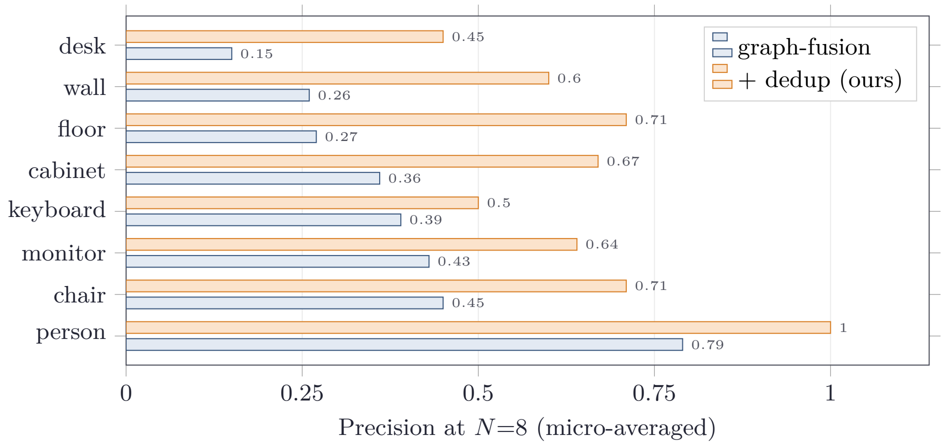

Fig. 7. Where the gain comes from: per-class precision at eight views for the over-counted classes (Table 6). Duplicate-aware fusion sharply recovers precision on the classes the lifted system over-counts (`desk` 0.15 → 0.45, `floor` 0.27 → 0.71, `monitor` 0.43 → 0.64) while leaving recall unchanged for almost every class. The error it fixes is over-counting of correctly-found objects, not mislabeling.

on 22–27 of 28 scenes, while a recalibrated, fully swept uncertainty-aware fusion gate gives no improvement at all and in fact regresses. These two results are not coincidental: they are two sides of the same mechanism. This section explains that mechanism, draws the methodological lesson from the circular evaluation that previously hid it, and is candid about the limits of the present study.

6.1 Why duplicate-aware merging helps and uncertainty does not

The dominant error of the lifted system is *over-counting*: a single physical object survives cross-view fusion as several nodes. The per-class characterization makes this concrete. As the view count grows from three to ten, the system's recall rises (micro-averaged 0.671 → 0.808) but its precision falls in lockstep (0.533 → 0.322), so net F1 is best at the sparse end and degrades with more frames. The classes that drive the precision collapse are exactly the ones that are *always* found and then fragmented: `monitor`, `desk`, `keyboard`, and `wall` all reach recall ≈ 1.0 while their precision falls toward 0.2–0.3. More views supply more correct detections of the same instance, and 3D fusion fails to collapse them. The failure is over-segmentation of correct objects, not mislabeling.

Read on this axis, the two interventions are opposites. Duplicate-aware fusion merges same-label nodes whose axis-aligned 3D boxes overlap; it is a *merge*, and merging is the corrective move when the error is over-splitting. The per-class effect is exactly as predicted: at eight views, precision jumps from 0.38 to 1.00 on `monitor`, 0.19 to 0.83 on `desk`, and 0.31 to 0.62 on `keyboard`, with recall preserved for nearly every class. The net is a uniform F1 gain of +0.068, +0.089,

Table 6. Per-class mechanism of duplicate-aware fusion at v8 (micro-averaged over 28 scenes). Dedup recovers precision on the over-counted classes while preserving recall for almost all classes; the only recall cost is on closely-packed same-class instances (`desk`, `bottle`).

Class	GRAPH-FUSION P/R	dedup P/R	Δ recall
`monitor`	0.43 / 0.95	0.64 / 0.95	0.00
`chair`	0.45 / 0.91	0.71 / 0.91	0.00
`floor`	0.27 / 0.93	0.71 / 0.93	0.00
`wall`	0.26 / 0.69	0.60 / 0.69	0.00
`desk`	0.15 / 0.96	0.45 / 0.92	−0.04
`cabinet`	0.36 / 0.27	0.67 / 0.27	0.00
`keyboard`	0.39 / 1.00	0.50 / 1.00	0.00
`person`	0.79 / 0.79	1.00 / 0.79	0.00
`bottle`	0.70 / 0.76	0.78 / 0.72	−0.04

+0.145, and +0.146 over the strongest non-merging fusion baseline at three, five, eight, and ten views, and it is the only variant whose accuracy *rises* as more views are added — because each added view supplies redundancy that the merge can absorb rather than fragmentation that it cannot.

The uncertainty-aware gate acts on the same axis but in the wrong direction. It *tightens* the merge criterion for high-uncertainty pairs, i.e. it splits more readily exactly where the system already over-splits. Against the independent reference it never helps: across the entire weight sweep $w \in [0.1, 0.8]$ no setting beats the non-gated baseline, and the wired setting regresses by −0.033, −0.053, −0.082, and −0.096 across the four view counts, losing on essentially all 28 scenes at five, eight, and ten views ($p < 0.001$). This is not a tuning failure to be patched away; it is the predictable consequence of suppressing merges in a regime whose error is too few merges. We report it as a feature of the study, not an embarrassment: knowing which lever points the wrong way is what lets us identify the one that points the right way, and the contrast between the two variants is the clearest evidence for the over-counting diagnosis itself.

6.2 The circular-evaluation lesson

This mechanism was invisible under the evaluation we — like much of this literature — started with. When the reference graph is seeded from the model's own dense-view prediction, the protocol is circular: a fusion method that simply reproduces its ten-view output scores 1.0 at the reference view *by construction*, and recall-raising behavior is rewarded regardless of whether it corresponds to real objects. Under that prediction-seeded reference the uncertainty gate appeared to be a sparse-view win (+0.040 and +0.021 at three and five views) and the over-counting precision sink was masked by the artificial dense-view ceiling. Regrading the identical model outputs against an *independently authored* human reference — written from the frames, never from the predictions — reverses the

Table 7. IoU-threshold sweep for duplicate-aware fusion (object-label F1, mean over 28 scenes). The gain over GRAPH-FUSION is robust across the low-IoU band $[0, 0.2]$ and only fades for strict thresholds. Default τ=0.1 in bold.

IoU τ	v3	v5	v8	v10	avg Δ
0.0 (any overlap)	0.581	0.593	0.610	0.605	+0.118
0.05	0.573	0.576	0.590	0.582	+0.101
0.1 (default)	**0.570**	**0.575**	**0.585**	**0.573**	+0.096
0.15	0.565	0.568	0.573	0.561	+0.087
0.2	0.558	0.557	0.560	0.546	+0.075
0.3	0.544	0.535	0.534	0.522	+0.054
0.5	0.529	0.517	0.503	0.490	+0.030

sign of the uncertainty result and removes the dense-view artifact, exposing the precision collapse and the merging fix underneath.

The methodological takeaway generalizes beyond this system. A reference seeded from any single model's output does not measure recognition; it measures agreement with that model, and it silently grants a perfect score to whatever produced the seed. In sparse-view 3D scene understanding, where dense ground truth is expensive and the temptation to bootstrap labels from a strong model is real, this failure mode is easy to fall into and hard to notice, because the inflated numbers look like success. We therefore treat the independent, human-verified reference and the documentation of the circular-pseudo-reference pitfall as a first-class contribution, and we recommend that any prediction-seeded reference be reported alongside an independently authored one whenever the latter is feasible. Concerns about evaluation construction and reproducibility in 3D scene understanding have been raised before [5,7]; our point is the specific and easily overlooked case where the reference is derived from the system under test.

6.3 Limitations

Evaluation scope and reference provenance. The benchmark spans 28 TUM RGB-D sequences, of which 5 (the paper-subset gold core) are human-verified and 23 carry two-pass VLM-drafted references (draft + adversarial verify), pending human verification. Both main findings reach $p < 0.001$ over the 28 scenes and also hold on the 5 human-verified scenes alone (+0.068 to +0.146 dedup gain), so they do not rest on the machine-drafted labels. Fully human-verifying the 23 drafted references is nonetheless the natural next step to remove any residual labeling bias, and the benchmark remains modest relative to large posed-RGB-D datasets.

Closely-packed same-class instances. Box-IoU merging cannot separate distinct instances of the same class whose axis-aligned 3D boxes overlap. A desk holding two or three monitors side by side is the failure case: the merge collapses them into one node, costing -0.33 recall on `monitor` and -0.17 on `desk` at eight views. The net effect is still strongly positive because the precision gain dwarfs the recall cost, but geometry alone is the wrong tool for telling two adjacent same-class objects apart, and this is the one place the merge does harm.

Static, "thing"-centric, rule-based. The pipeline assumes a static incoherently. "Stuff" classes such as `wall`, `floor`, and `ceiling` are treated as countable objects, which is a poor fit for amorphous surfaces and inflates the over-counting they exhibit; they could reasonably be reported separately or handled by a region rather than an instance model. Finally, the spatial relations are inferred from centroid and distance rules. These are inspectable and easy to audit, but they cannot express the support, containment, functional, or activity relations studied in richer 3D scene graph settings [1,25,4].

6.4 Future work

The cleanest next step follows directly from the merge's one failure mode. Box-IoU over-merges closely-packed same-class instances; appearance- or feature-based instance separation — splitting a merged group back apart when its members are visually distinct — is the natural remedy. This is also, notably, where the recall-oriented signal that the uncertainty gate tried and failed to add could finally pay off: not as a *merge gate* that prevents fusion in the over-counting regime, but as a *split decision* applied to an already-merged group, exactly the direction in which a recall signal helps rather than hurts. The negative uncertainty result thus does not retire the idea; it relocates it to the one operation where it is mechanistically appropriate. Beyond this, replacing the rule-based predicates with a learned relation module — once the independently authored annotations provide sufficient object-relation supervision — and reporting "stuff" classes under a region model are the obvious extensions.

7 Conclusion

We presented a detector-grounded open-vocabulary 3D scene-graph system that turns a handful of casual RGB views into an inspectable, language-grounded object-relation graph with all heavy models frozen and no per-scene optimization: an open-vocabulary detector (OWLv2) [15] proposes labeled boxes, feed-forward VGGT [26] geometry lifts them to 3D, and a cross-view fusion step assembles the graph. On an independent, human-verified TUM RGB-D [23] benchmark we showed three things. The dominant error is over-counting, and a one-line duplicate-aware merge corrects it, improving object-label F1 by $+0.06$ to $+0.12$ at every view count and winning on 22–27 of the 28 scenes ($p < 0.001$). A recalibrated, fully swept uncertainty-aware fusion gate — the move in the opposite direction — gives no benefit, a negative result that a circular, prediction-seeded reference had previously disguised as a win. And the de-circularized reference that exposed both, together with the documented pitfall that produced the original illusion, is the methodological core we most hope outlasts this particular system. What actually helps in sparse-view 3D scene graphs, on honest labels, turned out to be the simplest move pointed in the right direction.

References

1. Armeni, I., He, Z.Y., Gwak, J., Zamir, A.R., Fischer, M., Malik, J., Savarese, S.: 3d scene graph: A structure for unified semantics, 3d space, and camera. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 5664–5673 (2019)
2. Campos, C., Elvira, R., Gomez Rodriguez, J.J., Montiel, J.M.M., Tardos, J.D.: ORB-SLAM3: An accurate open-source library for visual, visual-inertial, and multimap SLAM. IEEE Transactions on Robotics **37**(6), 1874–1890 (2021)
3. Dai, A., Chang, A.X., Savva, M., Halber, M., Funkhouser, T., Niessner, M.: ScanNet: Richly-annotated 3d reconstructions of indoor scenes. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 5828–5839 (2017)
4. Gu, Q., Kuwajerwala, A., Morin, S., Jatavallabhula, K.M., Sen, B., Agarwal, A., Rivera, C., Paul, W., Ellis, K., Chellappa, R., Gan, C., de Melo, C.M., Tenenbaum, J.B., Torralba, A., Shkurti, F., Paull, L.: ConceptGraphs: Open-vocabulary 3d scene graphs for perception and planning. In: Proceedings of the IEEE International Conference on Robotics and Automation (ICRA). pp. 5021–5028 (2024)
5. Gundersen, O.E., Kjensmo, S.: State of the art: Reproducibility in artificial intelligence. In: Proceedings of the AAAI Conference on Artificial Intelligence. pp. 1644–1651 (2018)
6. Jatavallabhula, K.M., Kuwajerwala, A., Gu, Q., Omama, M., Chen, T., Li, S., Iyer, G., Saryazdi, S., Keetha, N., Tewari, A., Tenenbaum, J.B., de Melo, C.M., Krishna, M., Paull, L., Shkurti, F., Torralba, A.: ConceptFusion: Open-set multimodal 3d mapping. Robotics: Science and Systems (RSS) (2023)
7. Kapoor, S., Narayanan, A.: Leakage and the reproducibility crisis in machine-learning-based science. Patterns **4**(9), 100804 (2023)
8. Kerbl, B., Kopanas, G., Leimkuehler, T., Drettakis, G.: 3d gaussian splatting for real-time radiance field rendering. ACM Transactions on Graphics **42**(4) (2023)
9. Kerr, J., Kim, C.M., Goldberg, K., Kanazawa, A., Tancik, M.: LERF: Language embedded radiance fields. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 19729–19739 (2023)
10. Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.Y., Dollar, P., Girshick, R.: Segment anything. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 4015–4026 (2023)
11. Leroy, V., Cabon, Y., Revaud, J.: Grounding image matching in 3d with MASt3R. In: Computer Vision – ECCV 2024. pp. 71–91 (2024)
12. Li, J., Saltori, C., Poiesi, F., Sebe, N.: Cross-modal and uncertainty-aware agglomeration for open-vocabulary 3d scene understanding. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 19390–19400 (2025)

13. Liu, S., Zeng, Z., Ren, T., Li, F., Zhang, H., Yang, J., Jiang, Q., Li, C., Yang, J., Su, H., Zhu, J., Zhang, L.: Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In: Computer Vision – ECCV 2024. pp. 38–55 (2024)
14. Mildenhall, B., Srinivasan, P.P., Tancik, M., Barron, J.T., Ramamoorthi, R., Ng, R.: NeRF: Representing scenes as neural radiance fields for view synthesis. In: Proceedings of the European Conference on Computer Vision (ECCV). pp. 405–421 (2020)
15. Minderer, M., Gritsenko, A., Houlsby, N.: Scaling open-vocabulary object detection. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 72983–73007 (2023)
16. Minderer, M., Gritsenko, A., Stone, A., Neumann, M., Weissenborn, D., Dosovitskiy, A., Mahendran, A., Arnab, A., Dehghani, M., Shen, Z., Wang, X., Zhai, X., Kipf, T., Houlsby, N.: Simple open-vocabulary object detection with vision transformers. In: Computer Vision – ECCV 2022. pp. 728–755 (2022)
17. Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., Assran, M., Ballas, N., Galuba, W., Howes, R., Huang, P.Y., Li, S.W., Misra, I., Rabbat, M., Sharma, V., Synnaeve, G., Xu, H., Jegou, H., Mairal, J., Labatut, P., Joulin, A., Bojanowski, P.: DINOv2: Learning robust visual features without supervision. arXiv preprint arXiv:2304.07193 (2023)
18. Peng, S., Genova, K., Jiang, C.M., Tagliasacchi, A., Pollefeys, M., Funkhouser, T.: OpenScene: 3d scene understanding with open vocabularies. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 815–824 (2023)
19. Piekenbrinck, J., Schmidt, C., Hermans, A., Vaskevicius, N., Linder, T., Leibe, B.: OpenSplat3D: Open-vocabulary 3d instance segmentation using gaussian splatting. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). pp. 5285–5294 (2025)
20. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 8748–8763 (2021)
21. Schonberger, J.L., Frahm, J.M.: Structure-from-motion revisited. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4104–4113 (2016)
22. Straub, J., Whelan, T., Ma, L., Chen, Y., Wijmans, E., Green, S., Engel, J.J., Mur-Artal, R., Ren, C., Verma, S., Clarkson, A., Yan, M., Budge, B., Yan, Y., Pan, X., Yon, J., Zou, Y., Leon, K., Carter, N., Briales, J., Gillingham, T., Mueggler, E., Pesqueira, L., Savva, M., Batra, D., Strasdat, H.M., De Nardi, R., Goesele, M., Lovegrove, S., Newcombe, R.: The Replica dataset: A digital replica of indoor spaces. arXiv preprint arXiv:1906.05797 (2019)
23. Sturm, J., Engelhard, N., Endres, F., Burgard, W., Cremers, D.: A benchmark for the evaluation of RGB-D SLAM systems. In: Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). pp. 573–580 (2012)
24. Takmaz, A., Fedele, E., Sumner, R., Pollefeys, M., Tombari, F., Engelmann, F.: OpenMask3D: Open-vocabulary 3d instance segmentation. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 68367–68390 (2023)

25. Wald, J., Dhamo, H., Navab, N., Tombari, F.: Learning 3d semantic scene graphs from 3d indoor reconstructions. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 3961–3970 (2020)
26. Wang, J., Chen, M., Karaev, N., Vedaldi, A., Rupprecht, C., Novotny, D.: VGGT: Visual geometry grounded transformer. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 5294–5306 (2025)
27. Wang, S., Leroy, V., Cabon, Y., Chidlovskii, B., Revaud, J.: DUSt3R: Geometric 3d vision made easy. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 20697–20709 (2024)
28. Zhou, X., Girdhar, R., Joulin, A., Krähenbühl, P., Misra, I.: Detecting twenty-thousand classes using image-level supervision. In: Computer Vision – ECCV 2022. pp. 350–368 (2022)